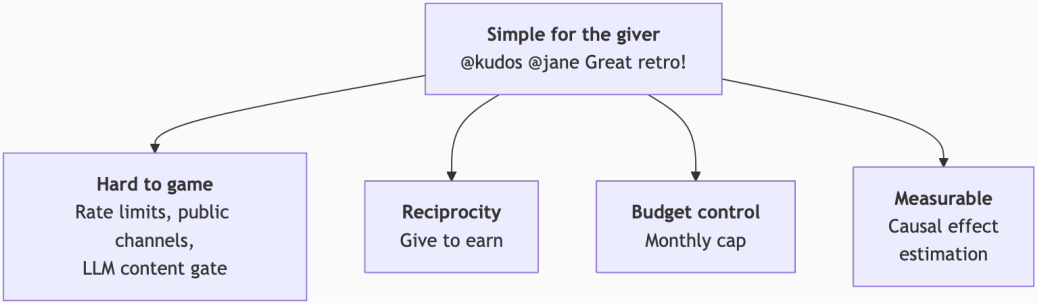
Kudos Bot


Simple for the giver
@kudos @jane Great retro!
Hard to game
Rate limits, public channels, LLM content gate
Reciprocity
Give to earn
Budget control
Monthly cap
Measurable
Causal effect estimation

Reciprocity: You Earn by Giving

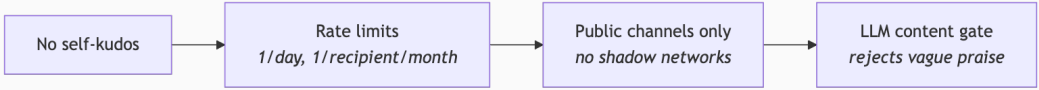
Your Nth give redeems your Nth received kudos — paired 1:1, oldest first. Giving kudos is not just altruistic — it's how you unlock your own earnings.

$$\text{owed} = \min(\text{given},\ \text{received}) - \text{redeemed}$$

	Given	Received	Redeemed	Owed
Alice	5	3	2	1
Bob	1	8	1	0

Bob has 7 unredeemed kudos. He earns his next payout by recognizing someone else.

Anti-Abuse: Defense in Depth



- Rate limits and public channels prevent reciprocal farming
- LLM gate produces a written record that praise was substantive — if a claim is false, the org won't be liable

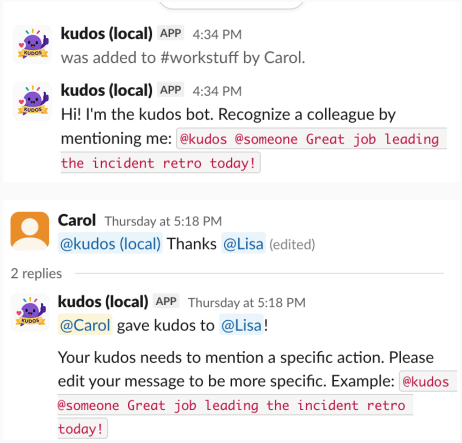
Budget Control

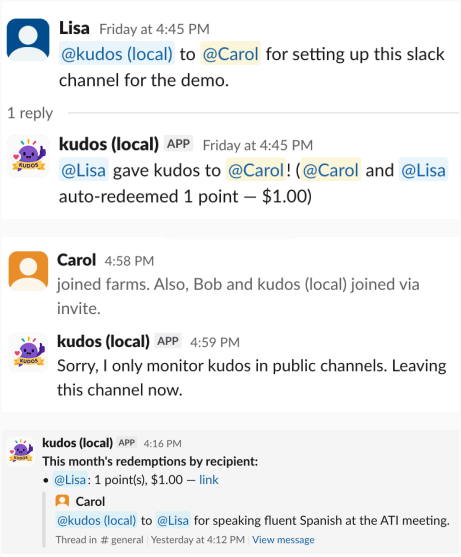
Monthly point budget and conversion rate set by accounting. Over-budget kudos are marked as overflow — payout opportunity lost.

	Time	Given	Received	Budget	Redemption
Alice Gave	10:00	2	2	3	1
Bob Received	10:00	3	3	2	1
Alice Gave	11:00	3	3	1	1
Bob Received	11:00	4	4	0	0

Bob's second receipt arrived after the budget was exhausted — it overflows and he earns nothing despite otherwise being owed.

Screenshots



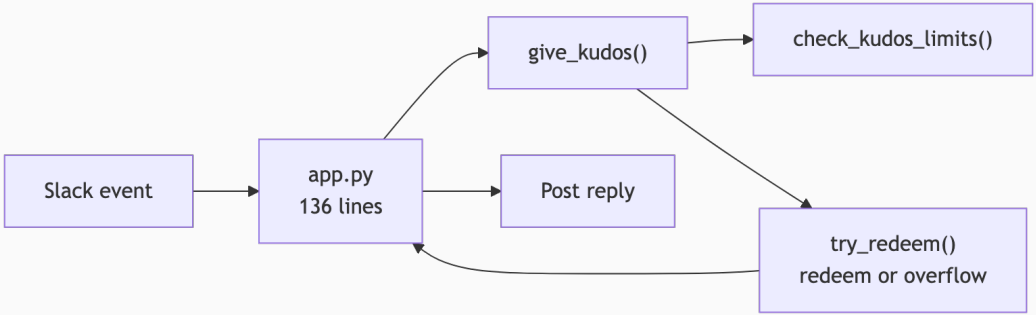


Dashboard Demo

Live demo: operational snapshot, usage & budget forecast, treatment effect plot, leaderboard, and topic drill-down.

Architecture

All business logic lives in Postgres functions. The Python app is a 136-line event router. 33 automated tests cover all business logic.



Edits delete the old kudos (un-redeeming linked points) and re-evaluate from scratch.

Scheduled Jobs

Script	Frequency	Purpose
`accounting.py`	Monthly	Report this month's redemptions to accounting channel
`weekly_reminder.py`	Weekly	DM users who haven't given kudos
`backfill.py`	Weekly	Embed kudos, cluster, LLM-summarize topics
`record_users.py`	Weekly	Record exposures for Poisson GLM

Treatment Effect Estimation

- Pairwise IRR between consecutive conversion-rate periods
- Kudos counts Y_j, exposures $E_j = \sum(\text{workday\_frac} \times \text{num\_users})$

$$\text{IRR} = \frac{Y_2/E_2}{Y_1/E_1} \qquad \text{CI via binomial test inversion}$$

- 90% confidence intervals on each IRR
- Forecast: Poisson prediction scaled by next week's exposure

Topic Clustering

Kudos messages are embedded into 128-dim vectors using a truncated embedding model, then clustered with KMeans using inverse-log month-frequency weights so older high-volume months don't dominate.

$$w_i = \frac{1}{\ln(1 + c_{m_i})} \qquad k = n_{\text{months}} + 3$$

Representative messages (nearest 25% to centroid) are sampled and summarized by an LLM into topic labels. Managers can see what behaviors are being recognized and how themes shift over time.

Demo: Diagnostics Notebook

Live demo: Poisson model diagnostics (quantile residuals, overdispersion, autocorrelation) and cluster diagnostics (elbow plot, silhouette scores, stability).

Technology Stack

Slack

Postgres

Python

Dash

statsmodels

scikit-learn

llama.cpp

Gemma

pg-schema-diff

Lines of Code

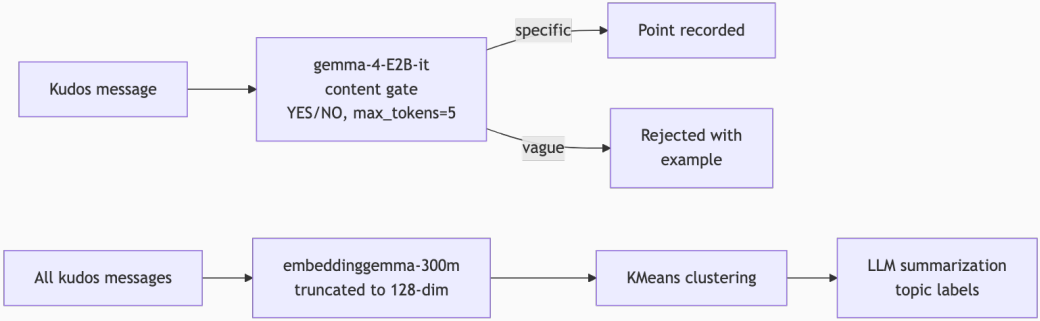
878 lines total — bot, dashboard, cron jobs, schema, and all business logic.

Component	Lines
Python	677
SQL	201
Total	**878**

AI in Development

- Critiquing the initial design
- Generating synthetic data (usernames, kudos messages, topic distributions)
- Prototyping all code and documentation (including this presentation)
- Tests and debugging
- Code review and critique

AI in the Product



The bot uses an LLM to gate every kudos for substantive content, and another to summarize topic clusters for the dashboard.

Deployment

Single Docker container runs `systemd` as PID 1 with everything inside.

Unit	Description
`postgresql.service`	Postgres with pgvector
`kudos-bot.service`	Slack bot via Socket Mode
`kudos-dashboard.service`	Gunicorn on port 8050
`kudos-backfill.timer`	Weekly embedding + clustering
`kudos-weekly-reminder.timer`	Weekly DM reminders
`kudos-accounting.timer`	Monthly redemption report

Schema migrations via `pg-schema-diff` — declarative SQL, no migration files.

Questions?

- One Slack message to give kudos — no forms, no approvals
- Anti-abuse enforced structurally, not by policy
- Reciprocity to encourage participation
- Causal measurement of budget impact on activity
- 845 lines of code, 33 tests including browser automations